

Junseo Lee – Curriculum Vitae

CONTACT

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RESEARCH INTERESTS

- Theoretical Physics: Quantum Many-Body Physics, Quantum Gravity, Bosonic Quantum Systems
- Theoretical Computer Science: Quantum Complexity, Algorithmic Statistics, Hardness of Approximation
- Quantum Information Science: Quantum Sensing, Quantum Metrology, Quantum Shannon Theory

EDUCATION

Harvard University, Kenneth C. Griffin Graduate School of Arts and Sciences (GSAS) Cambridge, MA
Doctor of Philosophy (Ph.D.) in Quantum Science and Engineering Sep. 2026 – Present

- Advisors: [Anurag Anshu](#) and [Sitan Chen](#) (Theoretical Computer Science); [Jordan Cotler](#) (Theoretical Physics)
- Affiliations: [Harvard Quantum Initiative](#); [Theory of Computation Group](#) (School of Engineering and Applied Sciences)
- Cross-registration at Massachusetts Institute of Technology (MIT), Fall 2026
- Supported by the Kwanjeong Doctoral Study Abroad Fellowship

Yonsei University Seoul, Korea
Bachelor of Science (B.S.) in Electrical and Electronic Engineering Mar. 2019 – Feb. 2023

- Thesis: *Combinatorial designs for information theory* (Advisor: Hong-Yeop Song)
- Supported by the Hyundai Motor CMK Science and Technology Scholarship

Chungnam Science High School Gongju, Korea
Concentration in Mathematics; Early Honors Graduate Mar. 2017 – Dec. 2018

RESEARCH POSITIONS

NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) Cambridge, MA
Junior Investigator (Nominator/Advisor: Jordan Cotler) May. 2026 – Present

Seoul National University (SNU) Seoul, Korea
Research Fellow, *Institute of Computer Technology* (Host: Kabgyun Jeong) Apr. 2026 – Aug. 2026
Research Affiliate, *Research Institute of Mathematics* (Host: Kabgyun Jeong) Jan. 2023 – Mar. 2026
Undergraduate Researcher, *Research Institute of Mathematics* (Advisor: Kabgyun Jeong) Mar. 2020 – Dec. 2022

Republic of Korea Army Seoul, Korea
Research Scientist (Technical Research Personnel), *Mandatory Military Service* Mar. 2023 – Mar. 2026

Yonsei University Seoul, Korea
Undergraduate Researcher, *High Dimensional Signal Processing Lab* (Advisor: Chulhee Lee) Jul. 2022 – Dec. 2022
Undergraduate Researcher, *Mathematical Biology Lab* (Advisor: Jeehyun Lee) Jan. 2022 – Jun. 2022

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea
Pre-Undergraduate Research Fellow, *Atomic-Scale Device Simulation Lab* (Advisor: Yong-Hoon Kim) Summer 2018
Pre-Undergraduate Research Fellow, *Therapeutic Biomaterials Engineering Lab* (Advisor: Ji-Ho Park) Summer 2017

PUBLICATIONS ([Google Scholar Profile](#))

*Equal contribution. †Authors listed alphabetically.

- (16) Myeongjin Shin, **Junseo Lee**, and Changhun Oh, “Heisenberg-limited Hamiltonian learning without short-time control,” (2026). [arXiv:2604.27838](#).
– Long talk at the [26th Asian Quantum Information Science Conference \(AQIS 2026\)](#).
- (15) Andreas Bluhm, Matthias C. Caro, Francisco Escudero Gutiérrez, **Junseo Lee**[†], Aadil Oufkir, Cambyse Rouzé, and Myeongjin Shin, “Certifying and learning local quantum Hamiltonians,” (2026). [arXiv:2603.29809](#).
– Extends and subsumes my earlier work [arXiv:2512.09778](#).
– Contributed talk at the [21st Conference on the Theory of Quantum Computation, Communication and Cryptography \(TQC 2026\)](#).
- (14) **Junseo Lee**[†] and Myeongjin Shin, “Optimal certification of constant-local Hamiltonians,” (2025). [arXiv:2512.09778](#).
- (13) Marco Fanizza, Vishnu Iyer, **Junseo Lee**[†], Antonio Anna Mele, and Francesco Anna Mele, “Efficient learning of bosonic Gaussian unitaries,” (2025). [arXiv:2510.05531](#).
– Contributed talk at the [29th Annual Conference on Quantum Information Processing \(QIP 2026\)](#).
- (12) Donghwa Ji, **Junseo Lee**, Myeongjin Shin, IlKwon Sohn, and Kabgyun Jeong, “Bounding quantum uncommon information with quantum neural estimators,” [Quantum Science and Technology](#) **11**, 015001 (2026). [DOI: 10.1088/2058-9565/ae18f4](#).
- (11) Kartik Anand, Kabgyun Jeong, and **Junseo Lee**[†], “Collapses in quantum-classical probabilistically checkable proofs and the quantum polynomial hierarchy,” (2025). [arXiv:2506.19792](#).
- (10) Myeongjin Shin*, **Junseo Lee***, Seungwoo Lee, and Kabgyun Jeong, “Resource-efficient algorithm for estimating the trace of quantum state powers,” [Quantum](#) **9**, 1832 (2025). [DOI: 10.22331/q-2025-08-27-1832](#).
- (9) Mingyu Lee, Myeongjin Shin, **Junseo Lee**, and Kabgyun Jeong, “Mutual information maximizing quantum generative adversarial networks,” [Scientific Reports](#) **15**, 32835 (2025). [DOI: 10.1038/s41598-025-18476-y](#).
- (8) Myeongjin Shin*, Seungwoo Lee*, **Junseo Lee***, Donghwa Ji, Hyeonjun Yeo, and Kabgyun Jeong, “Disentanglement provides a unified estimation for quantum entropies and distance measures,” [Physical Review A](#) **110**, 062418 (2024). [DOI: 10.1103/PhysRevA.110.062418](#).
- (7) Myeongjin Shin, **Junseo Lee**, and Kabgyun Jeong, “Estimating quantum mutual information through a quantum neural network,” [Quantum Information Processing](#) **23**, 57 (2024). [DOI: 10.1007/s11128-023-04253-1](#).
- (6) **Junseo Lee** and Kabgyun Jeong, “Quantum Rényi entropy functionals for bosonic gaussian systems,” [Physics Letters A](#) **490**, 129183 (2023). [DOI: 10.1016/j.physleta.2023.129183](#).
- (5) **Junseo Lee**, Hyeonjun Yeo, and Kabgyun Jeong, “Weighted p -Rényi entropy power inequality: Information theory to quantum Shannon theory,” [International Journal of Theoretical Physics](#) **62**, 253 (2023). [DOI: 10.1007/s10773-023-05512-8](#).
- (4) **Junseo Lee** and Kabgyun Jeong, “High-dimensional private quantum channels and regular polytopes,” [Communications in Physics](#) **31**, 189 (2021). [DOI: 10.15625/0868-3166/15762](#).
- (3) Kabgyun Jeong, **Junseo Lee**, *et al.*, “Single qubit private quantum channels and 3-dimensional regular polyhedra,” [New Physics: Sae Mulli](#) **68**, 232 (2018). [DOI: 10.3938/NPSM.68.232](#)
– Bronze Award, The Humantech Paper Award, Samsung Electronics (2018).

Book Chapters

- (2) **Junseo Lee**, “Assessing Quantum Integer Factorization Performance with Shor’s Algorithm,” In: *Quantum Computing: A Journey into the Next Frontier of Information and Communication Security*, Edited by Mohammad Hammoudeh, Abdullah T. Alessa, Amro M. Sherbeeni, Clinton M. Firth, Abdullah S. Alessa, [CRC Press](#) (2024). [DOI: 10.1201/9781003475286](#)

Patents

- (1) Kabgyun Jeong, Myeongjin Shin, and **Junseo Lee**, “Method for estimating quantum mutual information through a quantum neural network,” [Korea Patent Open No. 10-2026-0009068 \(2024\)](#). DOI: 10.8080/1020240091151

SELECTED HONORS AND AWARDS

Funding and Fellowships

- Doctoral Study Abroad Fellowship, *Kwanjeong Foundation*, 2026–2030
- Graduate Fellowship, *Harvard University*, 2026
- Academic Travel Grant, *Hyundai Motor CMK Foundation*, 2022
- Science and Technology Scholarship (full tuition and stipend), *Hyundai Motor CMK Foundation*, 2021–2022

Academic Excellence

- High Honors, *Yonsei University*, 2022
- Honors, *Yonsei University*, 2020–2021
- Science Scholarship, *JH Foundation*, 2017
- Academic Excellence Scholarship, *Asan Future Scholarship Foundation*, 2016

Additional Honors and Awards

- Selected Paper Award, Finance and Economics Contest, *DB Cultural Foundation*, 2022
- Best Tutor Award, *Yonsei University*, 2021–2022
- Third Prize, Undergraduate Research Exhibition, *Korean Physical Society*, 2021
- Bronze Award, Humantech Paper Award, *Samsung Electronics*, 2018
- Best Translator Award, *NAVER Connect Foundation & Khan Academy*, 2018
- Pre-Undergraduate Research Fellowship, *Korea Advanced Institute of Science and Technology*, 2017–2018
- Honorable Mention (National) & Gold Award (Regional), Korean Olympiad in Informatics (KOI), 2016

PROFESSIONAL ACTIVITIES

Journal Reviewer

- **American Physical Society:** Physical Review Letters, PRX Quantum, Physical Review Research, Physical Review Applied, Physical Review A
- **Nature Portfolio:** Nature Communications, npj Quantum Information
- **Institute of Electrical and Electronics Engineers:** IEEE Transactions on Information Theory
- **Others:** Quantum, Physics Letters A, Annalen der Physik, Mathematical Methods in the Applied Sciences, Statistics and Probability Letters

Conference Reviewer

- ACM-SIAM Symposium on Discrete Algorithms (SODA), 2027
- IEEE Symposium on Foundations of Computer Science (FOCS), 2026
- Theory of Quantum Computation, Communication and Cryptography (TQC), 2026
- Quantum Computing Theory in Practice (QCTiP), 2026
- Quantum Techniques in Machine Learning (QTML), 2025

Organizing

- Organizer, [Winter School on Quantum Learning Theory for Bosonic and Fermionic Systems](#), 2026
- Co-organizer, [Quantum Information Theory Seminar](#), *Seoul National University*, 2024–2025
- Co-organizer, [Quantum AI Hackathon](#), *Kakao Enterprise & Jeonju University*, 2025

Committee Service

- Poster Session Committee Member, National Undergraduate Quantum Conference, *Seoul National University*, 2026
- Selection Committee Member, [Quantum Internship Program](#), *National Information Society Agency & Korea Quantum Industry Center*, 2024 – 2025

Outreach and Community Resources

- Creator and Maintainer, [Quantum Learning Theory Zoo](#), 2025–present
- Facilitator (Mentor), Korea Scholar’s Conference for Youth (KSCY), *Yonsei University*, 2019

TEACHING

*Teaching materials available at: harris-junseo-lee.github.io. †University–Industry Research Internship.

- Organizer and Instructor, [Quantum Learning Theory for Bosonic and Fermionic Systems](#), *Quantum Information Science Club Association*, Winter 2026*
- Instructor, [Quantum Complexity Reading Group](#), *Quantum Information Science Club Association*, Fall 2025*
- Instructor, [Quantum Learning and Complexity Theory](#), *Quantum Information Science Club Association*, Summer 2025*
- (External) Instructor†, AAA559: College of Informatics Internship (II) (Graduate Course), *Korea University*, Fall 2025
- (External) Instructor†, AAA558: College of Informatics Internship (I) (Graduate Course), *Korea University*, Fall 2025
- (External) Instructor†, SW4343: Software Field Placement (I), *Korea Aerospace University*, Fall 2024
- Teaching Assistant, YCS1009: Change the World through Programming, *Yonsei University*, Fall 2022
- Teaching Assistant, YCS1002: Software Programming, *Yonsei University*, Fall 2022
- Teaching Assistant, EEE1108: Engineering Information Processing, *Yonsei University*, Fall 2021
- Course Tutor, MAT2016: Engineering Mathematics (III), *Yonsei University*, Spring 2022 (Best Tutor Award)
- Course Tutor, MAT1012: Engineering Mathematics (II), *Yonsei University*, Fall 2021 (Best Tutor Award)

RESEARCH VISITS

- Harvard University (funded by Anurag Anshu and the Harvard Quantum Initiative), Mar. 2026

SELECTED TALKS

*Online talk.

Research Talks

“Heisenberg-limited Hamiltonian learning without short-time control”

- Invited talk, *Technische Universität Hamburg (Hamburg University of Technology)*, Jun. 2026*
- Long contributed talk, 26th Asian Quantum Information Science Conference (**AQIS 2026**), *Korea Advanced Institute of Science and Technology (KAIST)*, Aug. 2026

“Certifying and learning local quantum Hamiltonians”

- Invited talk, Quantum Meets (Online Seminar Series), Jun. 2026*
- Invited talk, 7th QNRC Joint Workshop, *Korea Institute of Science and Technology Information (KISTI)*, May 2026
- Contributed talk, 21st Conference on the Theory of Quantum Computation, Communication and Cryptography (**TQC 2026**), *Delta Hotels Sherbrooke Conference Centre (Québec, Canada)*, Sep. 2026

“Optimal certification of constant-local Hamiltonians”

- Invited talk, Quantum Software Lab Seminar, *University of Edinburgh*, Mar. 2026*
- Contributed talk, Annual Meeting of the Quantum Information Society of Korea (QISK 2026), *Seoul SETEC Convention Center*, Feb. 2026

“Efficient learning of bosonic Gaussian unitaries”

- Invited talk, Annual Meeting of the Quantum Information Society of Korea (QISK 2026), *Seoul SETEC Convention Center*, Feb. 2026
- Invited talk, N³etFraST Workshop, *Korea Institute of Science and Technology Information (KISTI)*, Nov. 2025

- Invited talk, Quantum Data Science & AI Lab Seminar, *Yonsei University*, Nov. 2025
- Contributed talk, 29th Annual Conference on Quantum Information Processing (**QIP 2026**), *University of Latvia*, Jan. 2026 (presented under the title “Efficient Learning Algorithms for Structured Bosonic and Fermionic Unitary Operators”)

“Resource-efficient algorithm for estimating the trace of quantum state powers”

- Invited talk, *Electronics & Telecommunications Research Institute (ETRI)*, Dec. 2024
- Invited talk, Quantum Information Theory Seminar, *Seoul National University*, Dec. 2024*
- Invited talk, IBM-Yonsei Qiskit Fall Fest, *Yonsei University*, Nov. 2024*
- Invited talk, KISTI-KU-SNU Joint Workshop, *Seoul Biohub*, Oct. 2024
- Contributed talk, Annual Meeting of the Korean Mathematical Society (KMS), *Sungkyunkwan University*, Oct. 2024
- Poster, 28th Annual Conference on Quantum Information Processing (QIP 2025), *Raleigh Convention Center (NC, United States)*, Feb. 2025

“Mutual information maximizing quantum generative adversarial network”

- Invited talk, Triangle Quantum Computing Seminar, *NC State University Quantum Initiative*, Nov. 2023*

“Estimating quantum mutual information through a quantum neural network”

- Invited talk, CS Katha Barta, *National Institute of Science Education and Research (Bhubaneswar, India)*, Aug. 2023*

“Quantum Rényi entropy functionals for bosonic Gaussian systems”

- Poster, 25th Annual Conference on Quantum Information Processing (QIP 2022), *Caltech*, Mar. 2022

“High-dimensional private quantum channels and regular polytopes”

- Invited talk, KISTI-KU-SNU Joint Workshop, Sep. 2023*
- Invited talk, Quantum Information Theory Seminar, *Seoul National University*, Aug. 2021*
- Contributed talk, Winter Meeting of the Optical Society of Korea (OSK), *Daejeon Convention Center*, Feb. 2022
- Contributed talk, Fall Meeting of the Korean Physical Society (KPS), Oct. 2021*
- Poster, 25th Annual Conference on Quantum Information Processing (QIP 2022), *Caltech*, Mar. 2022

Tutorials and Public Lectures

“Getting Started in Quantum Information Science”

- Invited talk, Student Outreach Lecture, *Shinil High School (Seoul, Korea)*, Apr. 2026
- Invited talk, Student Outreach Lecture, *Shinil High School (Seoul, Korea)*, Aug. 2024

“Learning theory in ∞ -dimensional quantum systems”

- Invited talk, Team QST Summer Workshop, *Seoul National University*, Aug. 2025

“Introduction to quantum machine learning”

- Invited talk, Healthcare & Research Team Seminar, *Amazon Web Services (AWS Seoul, Korea)*, Mar. 2025

“Quantum machine learning models for drug library generation”

- Invited talk, Quantum Computing and Monte Carlo Workshop, *Yonsei University*, Aug. 2024

“QMA $\stackrel{?}{=} \text{NP}$: The NLTS theorem and the quantum PCP conjecture”

- Invited talk, Center for Quantum Network’s Channel Capacity Summer Workshop, *Seoul National University*, Jul. 2024

“Minimal data may be sufficient for quantum artificial intelligence”

- Invited talk, Department of Mathematical Sciences Seminar, *Seoul National University*, Jun. 2023*